Title: *Does removing race from eGFR change who an AI misses? Fairness of machine-learning screening for undiagnosed chronic kidney disease in NHANES 2017-2020*

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Background & Objective: Up to 90% of individuals with Chronic Kidney Disease (CKD) in the U.S. do not know they have it. As students exploring responsible AI in healthcare, our primary objective was to evaluate an internal validation of a two-stage machine learning framework for opportunistic digital screening in primary care, auditing fairness across race-adjusted 2009 and race-free 2021 eGFR equations.

Methods: Using the pre-pandemic NHANES 2017–March 2020 cohort (N = 15,560; adults aged ≥20 years), we defined undiagnosed CKD as eGFR < 60 mL/min/1.73m² or UACR ≥ 30 mg/g without prior diagnosis (KIQ022). Our Stage 1 model (EHR Zero-Lab) used non-invasive sociodemographic and vital variables for initial triage. Stage 2 (Lab-Enhanced) added routine blood biomarkers, strictly excluding creatinine and UACR to prevent target leakage. To study algorithmic fairness, we enforced demographic blindness by excluding race during training, reserving it for post-hoc fairness audits. We trained survey-weighted ElasticNet logistic regression and XGBoost models using a 70/30 stratified split, complex survey weights (WTMECPR), and population-weighted median imputation. Intersectional fairness was audited in low-income Black participants (Test N = 219, Cases N = 207) via complex survey bootstrap resampling (1,000 replicates) for False Negative Rate (FNR) and specificity.

Results: Regularized logistic regression provided a solid baseline (AUC = 0.782). Stage 1 Zero-Lab XGBoost achieved a weighted AUC = 0.735, while Stage 2 reached AUC = 0.780. At the Youden threshold (0.9143), survey bootstrap evaluation in the vulnerable subpopulation demonstrated a low FNR median of 11.11% (95% CI: 7.77%–14.29%) under both equations, with a median subgroup specificity of 33.33%.

Limitations & Conclusions: Main limitations include cross-sectional single-measurement data without 90-day KDIGO confirmation and estimated GFR. Nevertheless, our work shows that non-invasive, race-free machine learning can support equitable primary care screening, helping optimize referrals and catch silent CKD earlier in underserved communities.